

Mayank Garg

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WORK EXPERIENCE

Phemesoft – Project Intern (IoT & Cloud)

Jun 2024 – Jul 2024

- Contributed to the project by engineering the **cloud infrastructure on AWS** to support **smart irrigation system**, delivering a solution designed to optimize water usage leading to a projected **20% reduction** in water consumption.
- Developed and delivered the **user dashboard**, enabling real-time monitoring through **data visualization** for farmers.

GiftAble Foundation – Research Intern

Jun 2022 – Jul 2022

- Analyzed **20 years** of district-level data to **generate analytical snapshots** and report on economic trends.
- Evaluated policy impacts by analyzing multiple datasets to **provide actionable insights** and identify **5+ areas for strategic intervention**.
- Developed a national database of over **500 college placement cells in MS Excel**, improving the efficiency of partner and recruiter outreach efforts.

PROJECTS

SecurePixel – Creator & Developer

Feb 2023

- Engineered an **image encryption** system using **Henon map** iterations to **randomize pixel data**.
- Implemented a **steganography** layer to conceal the encrypted data within a cover image, ensuring high imperceptibility with a Peak Signal-to-Noise Ratio (PSNR) of **30+ dB**.
- Developed a **full-stack application** with a **Flask backend** and **HTML/CSS frontend**, leveraging OpenCV for core image encryption and processing tasks.

Smart Irrigation System – Developer

Jun 2024

- Architected and implemented a **serverless data pipeline** using **AWS Lambda** to process real-time IoT sensor data, configuring **AWS RDS** for data storage.
- Designed and deployed the supporting cloud infrastructure on **AWS**, enabling the project's machine learning model to achieve over **90% accuracy** by ensuring **reliable, real-time data ingestion**.
- Developed a **full-stack dashboard** using **React.js** for the frontend and **Node.js** for the backend, utilizing **AWS API Gateway** for request management.

Trendify – Creator & Developer

Feb 2024

Real-time Twitter sentiment analysis platform to **derive meaningful insights**.

- Developed a full-stack **sentiment analysis platform** using **FastAPI** and **React.js** that performed on-demand analysis based on user-submitted keywords and tweet quantities.
- Developed a **web scraping** pipeline with **Selenium** to fetch and parse the top tweets for any given keyword, with the entire end-to-end process handling requests for **100 tweets in under 60 seconds**.
- Implemented **VADER sentiment analysis** (NLTK) to classify tweets and designed a responsive **Tailwind CSS** interface to visualize sentiment breakdowns and representative tweets.

LEADERSHIP AND VOLUNTEERING EXPERIENCE

UPES Hypervision – Associate Technical Head

May 2023 - May 2024

Led and mentored a tech team, streamlined workflows, and ensured timely execution of high-impact events.

TECHNICAL SKILLS

Programming Languages : C++, Java, Python, SQL, JavaScript

Web Development: React.js, FastAPI, Node.js, Flask, HTML, CSS, Tailwind CSS

Cloud & DevOps : AWS, Google Cloud, Docker, Git

CERTIFICATIONS

- AWS Certified Developer – Associate

Sept 2024

EDUCATION

University of Petroleum and Energy Studies (UPES), Dehradun

Aug 2021 - May 2025

B.Tech. (Hons.) Computer Science, 2025 | CGPA: 7.66

Abhinav Public School, Delhi

April 2019 - March 2020

12th Grade, 2020 | Percentage : 75.8%

COMPETITIONS

- Secured **3rd place** in UPES Hypervision - **Code Hustle** for developing the Automated YouTube Video Tags Updater.
- Finalist** in 48-hour **UPES CSI Hackathon** for building an Image Encryption System.